Synthetic Biological Intelligence: Interfaces for In-Vitro Neuronal Learning

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Abstract. Demand for AI has grown exponentially in recent years and will continue to increase in humanity's pursuit of artificial general intelligence (AGI). Investors, institutions, and companies have poured billions of dollars into AI research in search of improved performance and energy efficiency. Looking towards the future, institutions and companies have started to explore alternative modalities for computation beyond the traditional semiconductor design, specifically quantum computing and biological computing. This paper explores the novel biological computation modality that draws inspiration from the world's most efficient computational model, the human brain. The field of biological computing remains widely unexplored, offering opportunities in the pursuit of energy-efficient computing. In this paper, we elucidate multiple methodologies and techniques focusing on training biological neural networks grown on micro-electrode arrays towards a goal-directed task. We also outline critical areas to explore in the future within the dynamic landscape of biological computing.

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1 Introduction

Investors, institutions, and companies have poured billions of dollars into the ever-expanding market of AI and machine learning in pursuit of this technological gold rush. As a result, the demand for energy-efficient computation modalities has risen sharply. In this technological era, there are two ways in which we can improve these modes of computation. First, we can explore improvements to current architectures. Second, we can explore new modalities for computation. In this paper, we focus on the latter approach. Quantum computing and biological computing have garnered significant interest due to their potential for scaling computation. Quantum computing leverages qubits to encode information but risks information loss due to quantum decoherence.

Biological computing, on the other hand, exploits the intricate network topology of the neural network to perform complex computation. It harnesses the power of biological neuronal networks (BNNs), which offer many potential applications and improvements over current AI and machine-learning paradigms. BNNs learn by modifying synaptic strengths through experience and forming neural patterns (memories) through adaptation, allowing neurons to be trained and to learn. Researchers currently build biological computers by plating neuronal cultures or organoids on a micro-electrode array (MEA). Once plated, they carefully tend to the cultures until they have matured enough to be trained.

Currently, there are two distinct training paradigms employed in BNNs: reservoir computing and reinforcement learning. The reservoir-computing approach primarily leverages the output of the living dynamical system and trains a downstream machine-learning model to interpret that output; it does not directly perturb the network to tune synaptic plasticity toward a favorable outcome. Therefore, in this work we focus on elucidating reinforcement-learning-based approaches. Before diving into the technical details of training a BNN, we first introduce the theoretical foundations of how a neuron learns.

2 Predictive Coding

Predictive coding is the theoretical foundation that grounds the training protocols described in this paper. [1,2] BNNs learn in a way that is fundamentally different from typical artificial neural networks (ANNs). ANNs are typically trained through a process called backpropagation, in which the gradient of a loss function is computed and propagated backward through the network to update the weights and minimize prediction error. Backpropagation is unlikely to exist in the brain because of temporal discontinuity and non-local plasticity. First, our brains process information in real time, making it highly implausible to have a separate backward pass for learning. Second, backpropagation requires downstream error signals to be available at local synapses, whereas neurons in the brain learn based only on local synaptic information.

Predictive coding addresses both of these concerns by reframing the model of the brain. In this framework, the brain creates an internal model of the world and projects a prediction μ down to its sensory neurons. Using this model, the brain computes a prediction error

$$\epsilon = x - \mu$$

which is the difference between what the brain actually senses and what it predicted it would sense. The system—in this case the brain—is efficient when the prediction error ϵ is equal to zero. When the prediction error is nonzero, the error is propagated upward into the brain's internal model of the world through a process that can be formalized in terms of variational free energy minimization. [1]

Mathematically, the predictive-coding model replaces backpropagation with the idea of minimizing a global energy (free-energy) function defined as the sum of weighted squared errors across all layers:

$$F = \sum_l \frac{1}{2} \epsilon_l^T \Pi \, \epsilon_l$$

where the error at layer l is the difference between the activity and the top-down prediction,

$$\epsilon_l = x_l - f(W x_{l+1})$$

The neural dynamics then follow gradient descent on this energy,

$$\dot{x}_l \propto -\frac{\partial F}{\partial x_l}$$

allowing the network to settle into a valid state, while synaptic weights are updated via a local Hebbian rule,

$$\Delta W_l \propto \epsilon_l \, f'(x_{l+1})^T$$

This model shows how biological neurons can approximate the powerful learning mechanisms in deep neural networks using only local signals at each synapse. [2]

3 Our Setup

3.1 Neural Cultures

Our biological computing setup builds upon previous iterations of bio-chips that have been successfully trained, including systems that learned to play Pong, CartPole, and Pac-Man. [3–5] The Pong paper used cortical cells from the dissected cortices of rodent embryos for their BNN culture. [3] These cultures, when maintained properly, survived for months when grown on MEAs and nurtured in nutrient-rich media. The CartPole paper successfully trained a BNN by using mouse cortical organoids in vitro derived from embryonic stem cells, using these organoids—rather than dissociated primary neurons—as the biological substrate for learning. [4] Similarly, the Pac-Man paper used paired brain organoids in a closed-loop setup to demonstrate long-term potentiation and task-specific learning. [5]

Due to the high variability, cost, and maintenance demands of organoids, we opted instead to use hippocampal place cells. Hippocampal place cells are neurons that fire at specific locations in an environment and are responsible for learning and memory. In contrast, cortical neurons more broadly refer to neocortical cells that support diverse sensory, motor, and cognitive functions.

3.2 Neural Culture Geometry

Most microelectrode array recording systems use flat, square devices whose electrode layout resembles the pin grid of a CPU socket on a motherboard. This footprint limits measurement to two dimensions. Three-dimensional cell cultures, however, provide a better representation of human physiology, and constraining these cultures to a two-dimensional recording setup does not fully capitalize on the rich environment a 3D culture presents. Both the Pong and CartPole implementations used flat neural preparations. [3, 4] Based on the success of these studies, using flat neural cultures does not appear to obstruct the learning process of biocomputers.

However, a 2D structural setup does not align with the 3D spatial organization of neurons in the brain. Emerging 3D designs in neural chips have multiple advantages over 2D designs because they more accurately capture the complexities of the nervous system from structural, functional, and effective-connectivity perspectives. [6] For this reason, we opted for a 3D approach in our neural-chip configuration. To mimic the multilayered organization of neural networks, we fabricated a biomaterial scaffold and folded it multiple times on top of itself to create an effectively 3D structure.

We used a biomaterial embedded with electrodes to allow both stimulation and recording. Specifically, we chose SU-8, an epoxy-based negative photoresist that has seen widespread adoption in biomedical applications due to its high biocompatibility in both in vivo and in vitro settings. [7]

3.3 Recording Channels

Our neural recording setups were limited by the number of available stimulation and recording sites on the channels of our MEA. Scaling from conventional 60-channel to 120-channel recordings in a 3D configuration required custom hardware development and significantly influenced the scope of our experiments.

4 Training Methodologies

The predictive-coding framework provides the theoretical foundation for explaining how our neural cultures can learn. [1, 2] We leverage the framework's drive to minimize prediction error to teach our neural cultures. Biology relies on local learning rules in which neurons adjust their local synaptic weights ω based on the activity of their immediate neighbors to reduce local prediction error. Mathematically, a simplified weight-update rule can be written as

$$\Delta w \propto \epsilon \cdot f(x)$$

where $f(x)$ denotes a postsynaptic activity term. Based on this framework, training protocols couple correct behaviors with predictable sensory feedback, while incorrect behaviors are associated with unpredictable, noisy input. This effectively forces neurons to reorganize their connectivity to avoid high-entropy, high-error states.

The Pong, CartPole, and Pac-Man systems all utilized this predictive-coding-inspired closed-loop framework. [3–5] In the Pong paper, after just five minutes of training, the neural culture demonstrated significant learning and improvement in its performance at playing Pong, providing evidence that neural networks tend to actively avoid high-entropy states. [3]

To train our cultures, we adopt the three-stage training method implemented in the CartPole paper. [4]

Stage 1: Identifying putative neurons. We first record the action potentials of the culture and identify units of activity with the highest signal fidelity. Signal fidelity is quantified as

$$\eta = (1 + \hat{r}) \cdot (1 + 0.1\hat{\mu}_{\text{amp}})$$

where $\hat{r}$ is the normalized log spike rate and $\hat{\mu}_{\text{amp}}$ is the mean spike amplitude. Electrodes with high η scores indicate healthy, active neurons. The identification of these high-activity channels creates a spatiotemporal footprint of the units present in our neural chip.

Stage 2: Mapping connectivity. In the second stage, we stimulate the putative neurons using biphasic voltage pulses. These stimulations allow us to estimate different orders

of connectivity in the culture. First-order connectivity (C_1) is defined as the probability of evoking a spike in a short-latency window following stimulation. When selecting neurons to stimulate during training, higher C_1 values correspond to ideal pathways. Higher-order connectivity levels (C_m) are characterized by average spike counts over longer latency windows.

Stage 3: Closed-loop training. In the third and final stage, we perform closed-loop training based on the task policy, which seeks to minimize a loss function derived from predictive-coding principles. The training method is closely analogous to reinforcement learning, where rewards and penalties are assigned according to the predictive-coding framework established earlier. By combining this framework with the three-stage methodology of recording, stimulating, and training, we aim to engineer optimal conditions for the neurons to learn.

5 Results

Although we developed a robust theoretical framework and implementation plan for training our cultures, our training and results were ultimately constrained by hardware and engineering limitations. We devoted substantial effort to devising and optimizing the training methodology, but could not fully overcome the limitations of our data-acquisition system.

In particular, scaling up the number of channels in our setup to record from and stimulate our 3D neural chip posed a major challenge. Our 3D culture design required expanding from 60 to 120 electrodes, which in turn required custom instrumentation. A significant portion of the independent work was spent designing and building this hardware.

Our 3D culture design also required several weeks to fabricate. When it came time for implementation, we were unable to reliably record action potentials from the 3D culture due to unresolved hardware issues. As a result, we pivoted to a 2D culture previously fabricated by the lab. Using this culture, we were able to record action potentials; however, our data-acquisition system still did not support the high-fidelity, low-latency stimulation and recording necessary for the closed-loop training protocols described above.

6 Conclusion and Future Considerations

Our theoretical understanding of closed-loop training in-vitro neural-culture through active inference and reinforcement learning forms the main focus of this thesis. Unfortunately, the experimental results were limited by both biological and engineering constraints. In particular, the design of a novel 3D microelectronic chip that seamlessly interfaces with commercial systems proved to be a major hurdle.

Nevertheless, we are confident that these hardware limitations are solvable. As improved instrumentation becomes available, the predictive-coding framework and training methodology detailed in this report will provide a roadmap for future BNN training experiments. Fully realizing such systems may enable scalable, energy-efficient biological computers that complement or augment conventional AI hardware.

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